

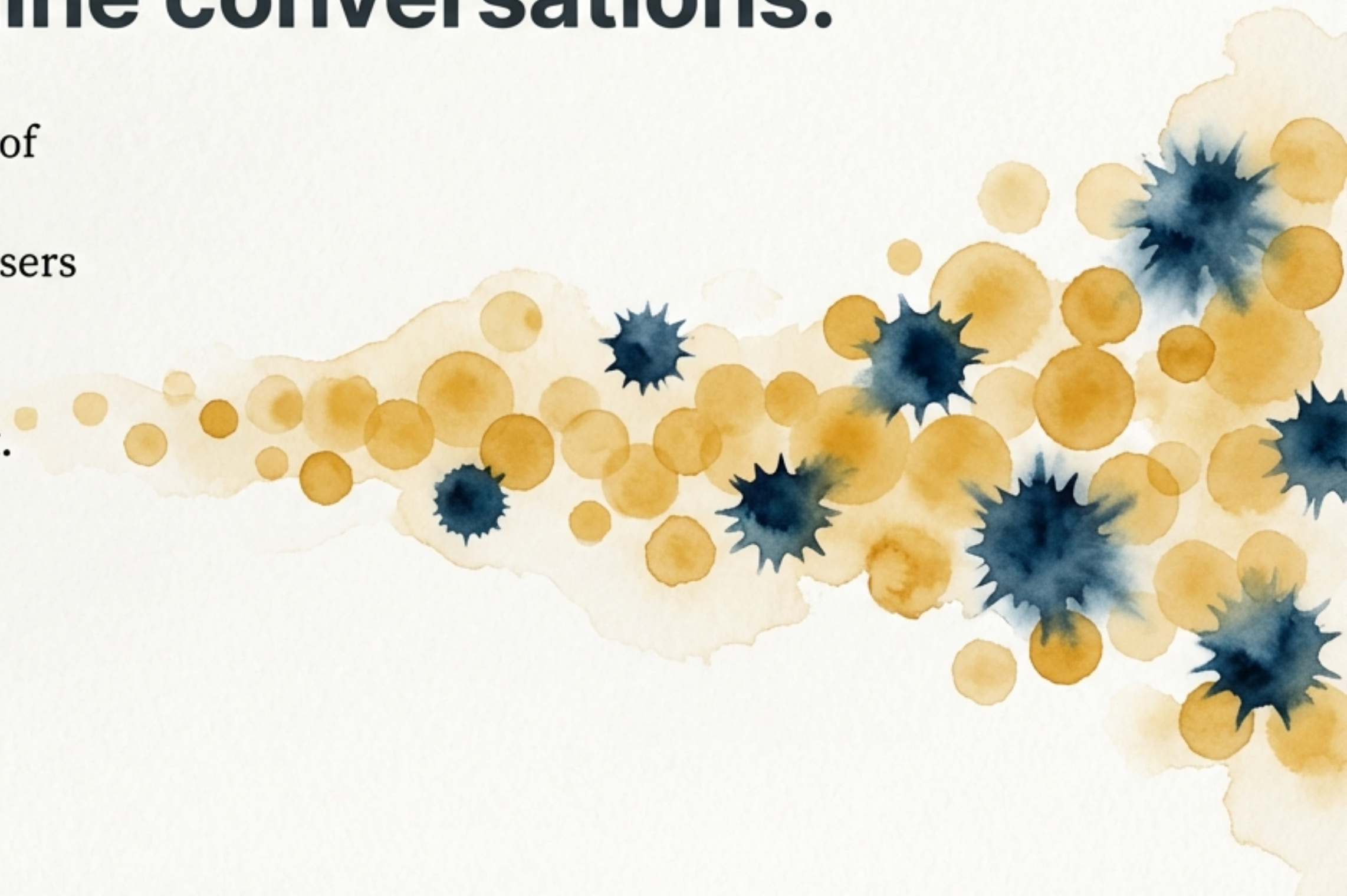
Crafting Clarity: A Labeler for Targeted Profanity on Bluesky



A project by Rachel Keels, Rohil Saraf, Jeffery Chen, and Po-Yo Juan.

The goal is not censorship, but user control over online conversations.

Targeted profanity pushes users out of conversations and creates hostile environments. Our approach gives users flexible tools to filter their own experience, choosing whether to hide, warn, or show profane content. This balances safety with freedom of expression.



Our system classifies posts into three categories, providing clear, actionable labels.



Targeted Profanity

Directed at a person or group
(e.g., a curse word + @mention
+ negative sentiment).



General Profanity

Pulled from a list of known
curse words used in a
non-targeted way.



Neutral

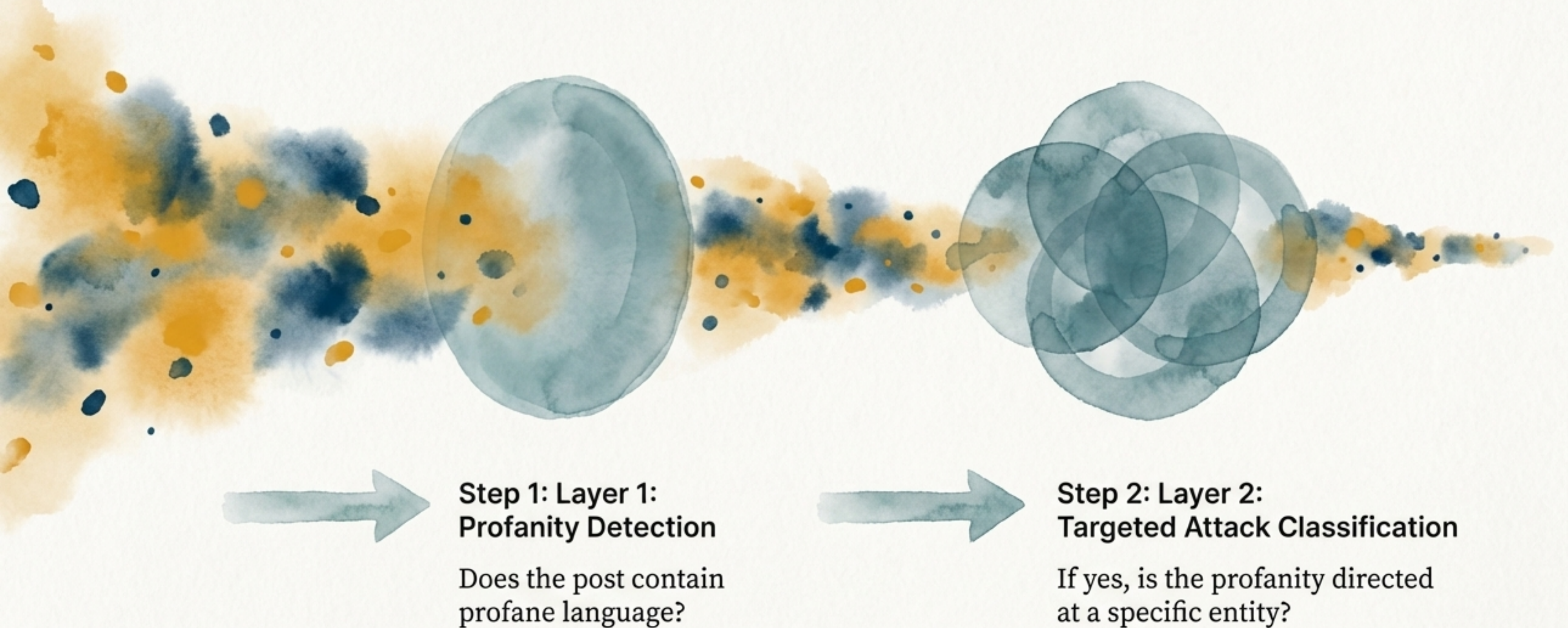
Posts without profanity.
No visible label is applied.

Hide

Warn

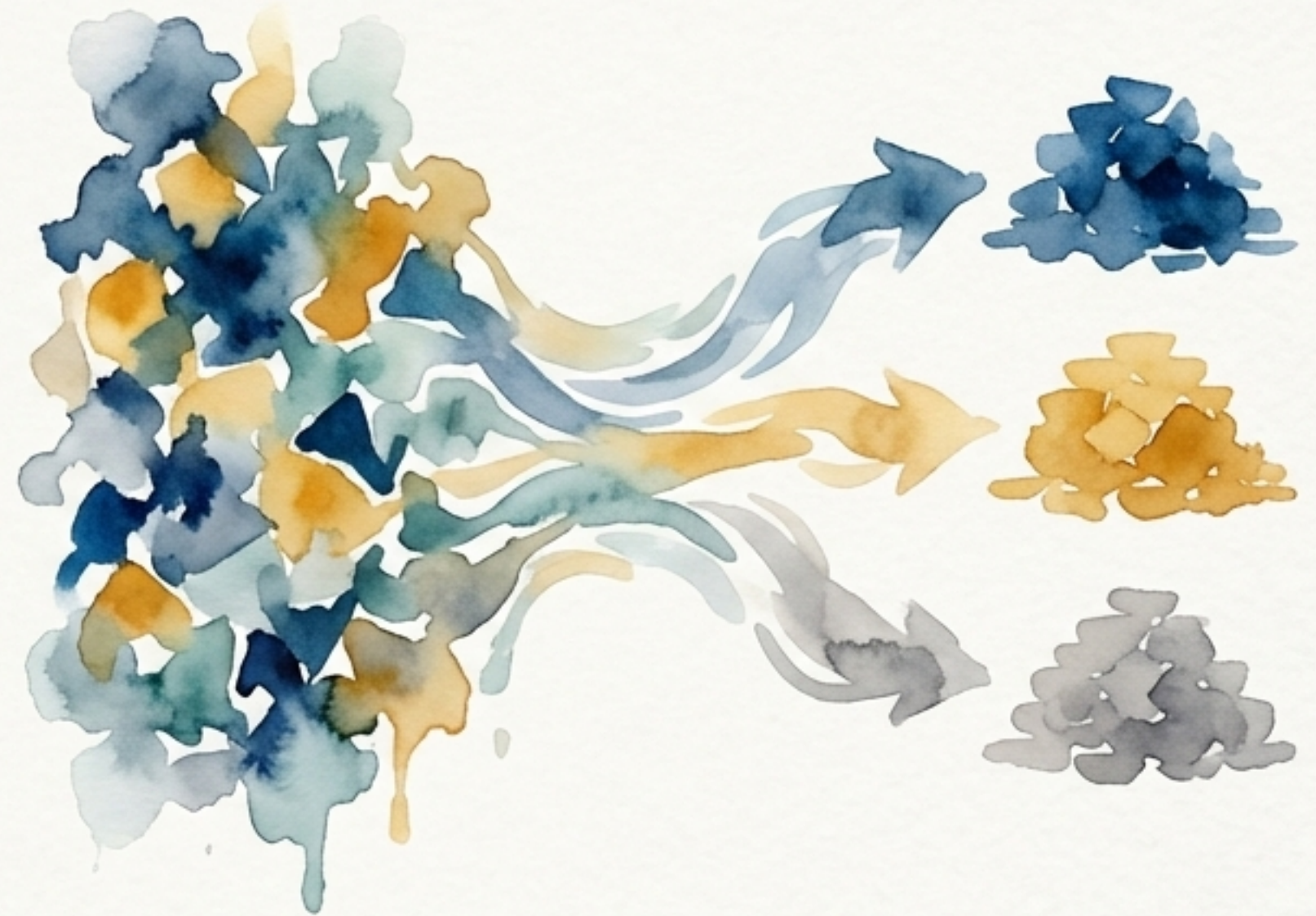
Show

We designed a two-layer pipeline to first find profanity, then identify if it's targeted.



Building a strong foundation started with rigorous data collection and annotation.

- **Collection***: Retrieved ~50 posts per keyword from `bad-words.txt` via the Bluesky API, including contextual data (quotes, mentions, replies).
- **Annotation***: Human-annotated a 150-post test set with a balanced class distribution. Used Llama3.1 to label a 5,000-post training set.
- **Key Insight***: Initial disagreements among annotators (e.g., 'Is slandering a movie 'targeted'?') forced us to develop more rigorous, consistent labeling definitions.



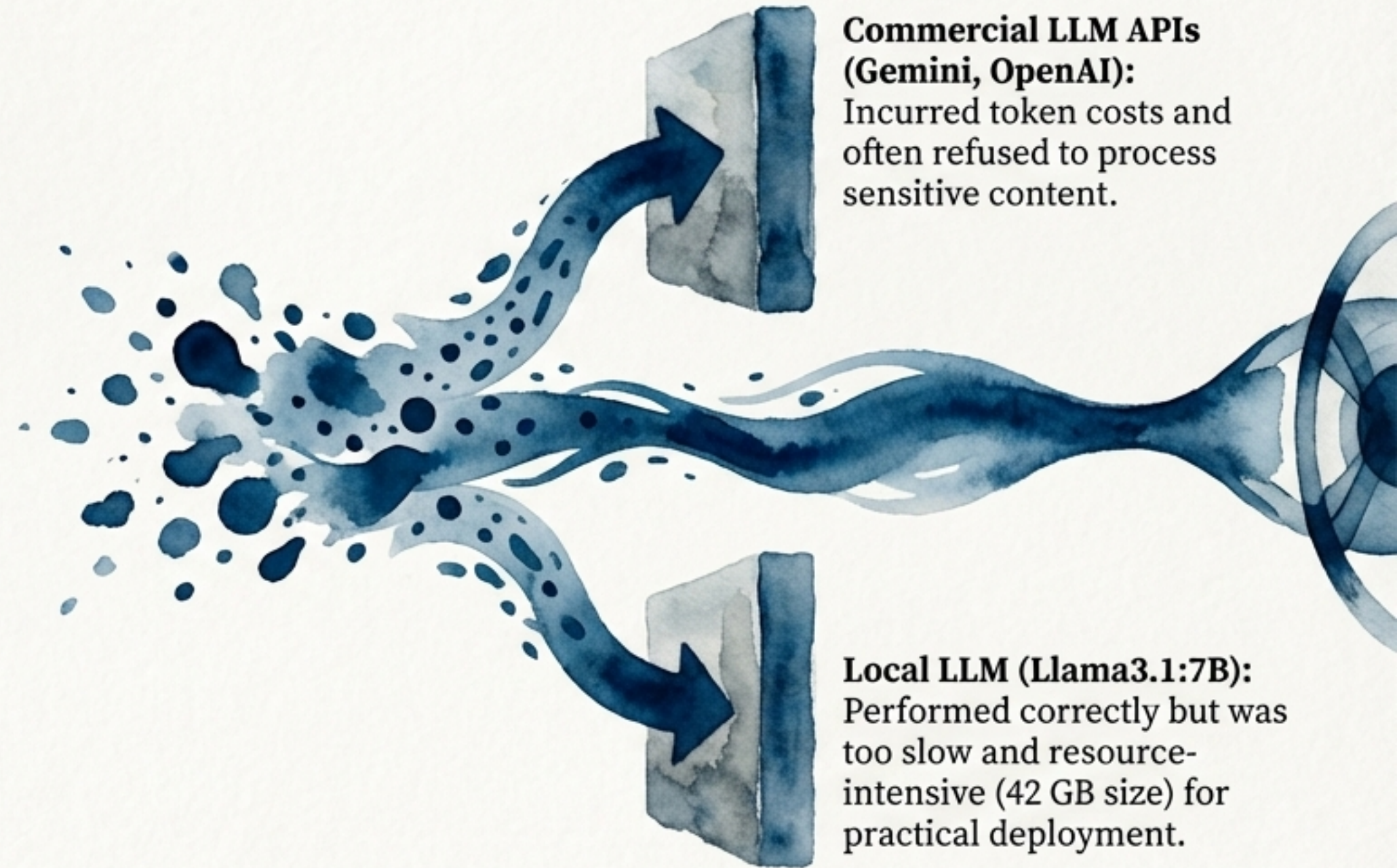
Simple sentiment analysis failed for profanity detection, leading us to a multi-model ensemble.

Initial tests with sentiment analysis produced excessive false positives, misclassifying sadness or depression as toxic. Our final solution combines three specialized models for comprehensive coverage.



Detecting targeted attacks is semantically complex; standard solutions were impractical.

Identifying intent and context is too abstract for simple classifiers. We explored large language models, but encountered significant roadblocks.



We used knowledge distillation to capture an LLM's power in a production-ready model.

We used the slow but powerful Llama3.1 as a "teacher" to train a smaller, faster "student" model. We chose `microsoft/deberta-v3-base` for its ability to handle complex semantic structures, like differentiating between a post and its quoted content.

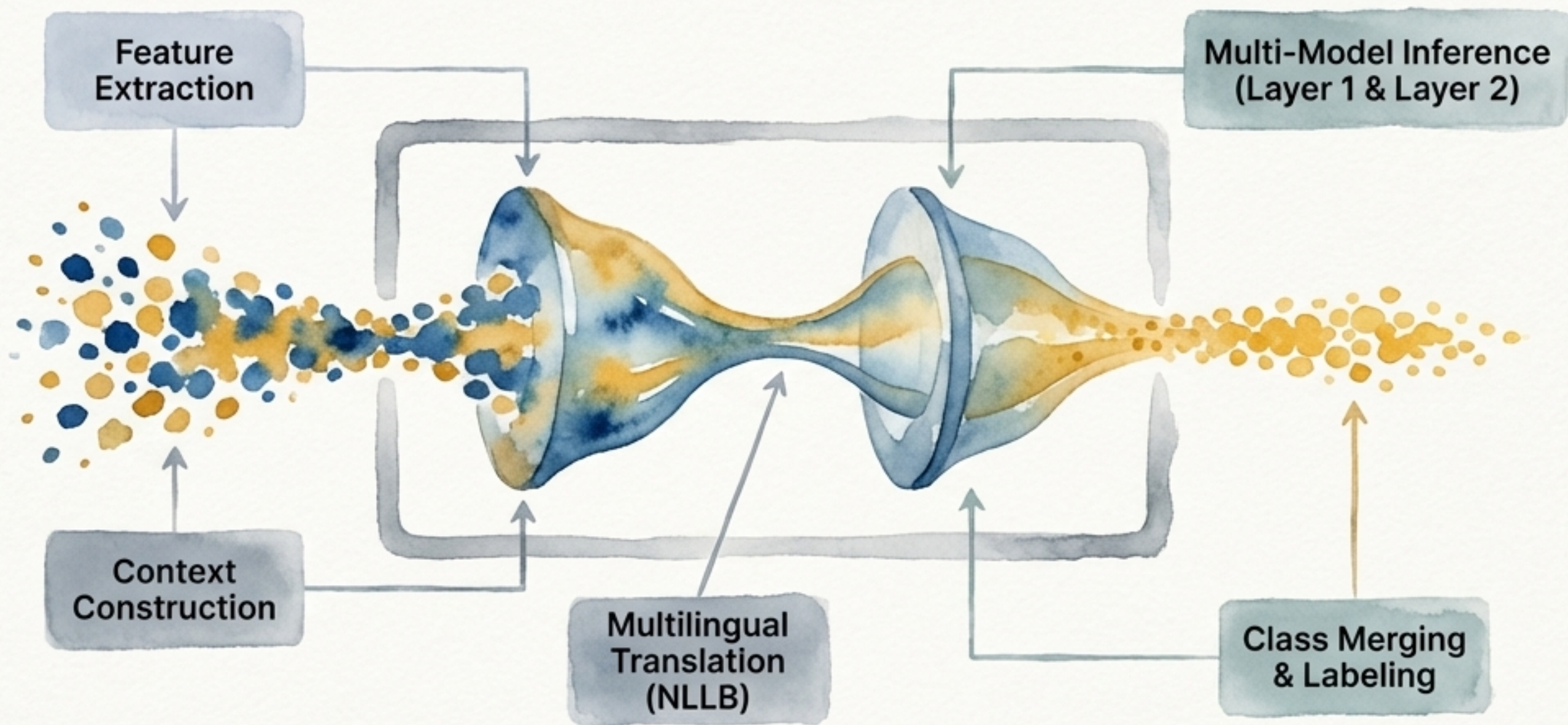


1. **Teacher:** Llama3.1 generated labels for the 5,000 training samples (~10 hours).

2. **Student:** DeBERTa-v3 was fine-tuned on these labels.

3. **Result:** The student model replicated the teacher's behavior with over 89% precision and 90% recall.

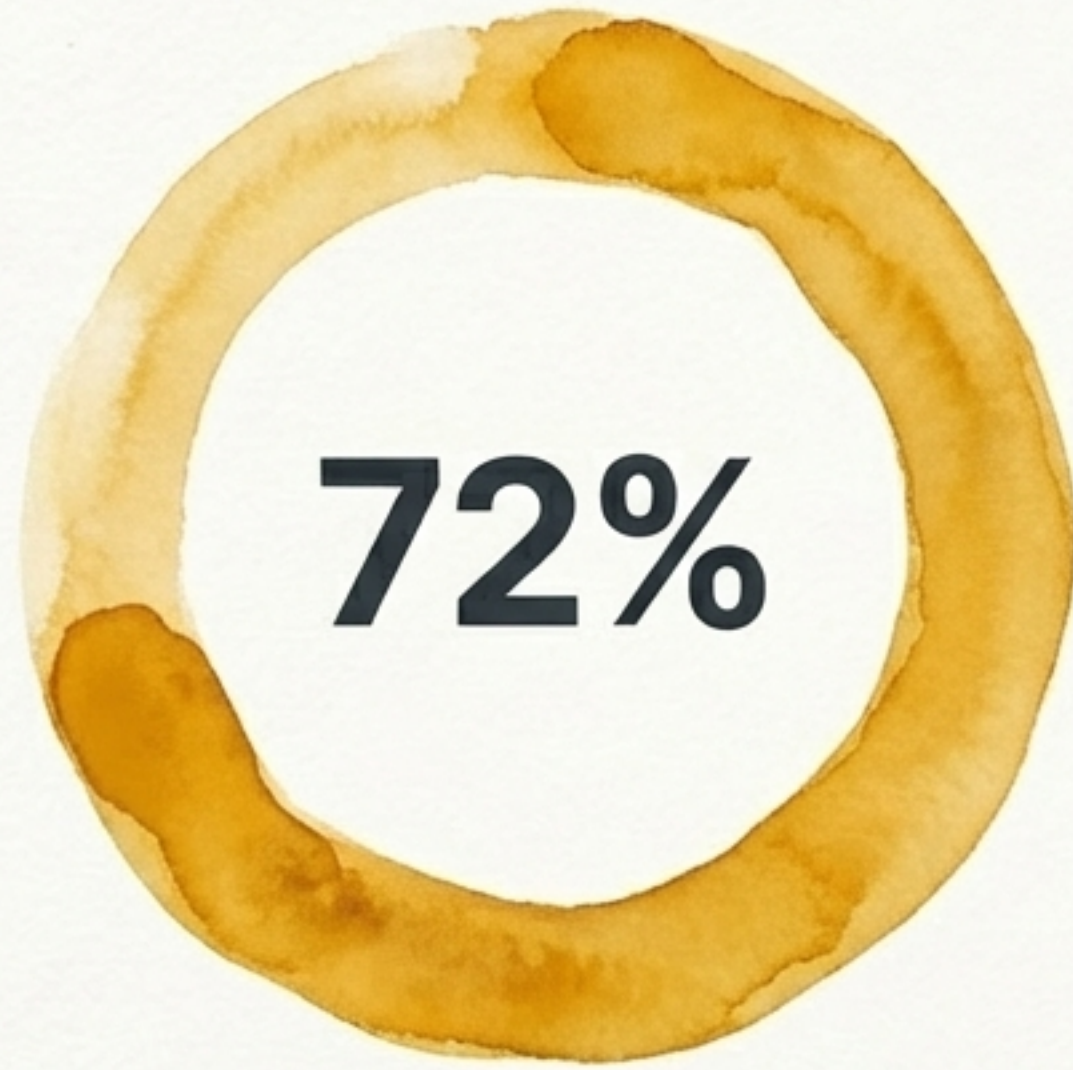
The final system is a fully modular, end-to-end pipeline.



****Key Takeaway*:** Each component can be independently tested and replaced, enabling future improvements and scalability.

The pipeline achieves 72% overall accuracy with strong performance in each layer.

Performance measured against the 150 manually annotated test samples.



Layer 1 (Profanity)



90% Precision / 88% Recall

Layer 2 (Targeted Attack)



81% Precision / 85% Recall

Computational performance is efficient enough for real-world deployment.

The complete pipeline processed the 150-sample test set in 104 seconds of wall-clock time.



**~0.7 seconds
per post**

****CPU Time****: 33.52s (user), 12.49s (system)

****CPU Utilization****: 44% average

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The system can scale without requiring excessive computational resources.

We prioritized high precision to build user trust and avoid excessive false positives.

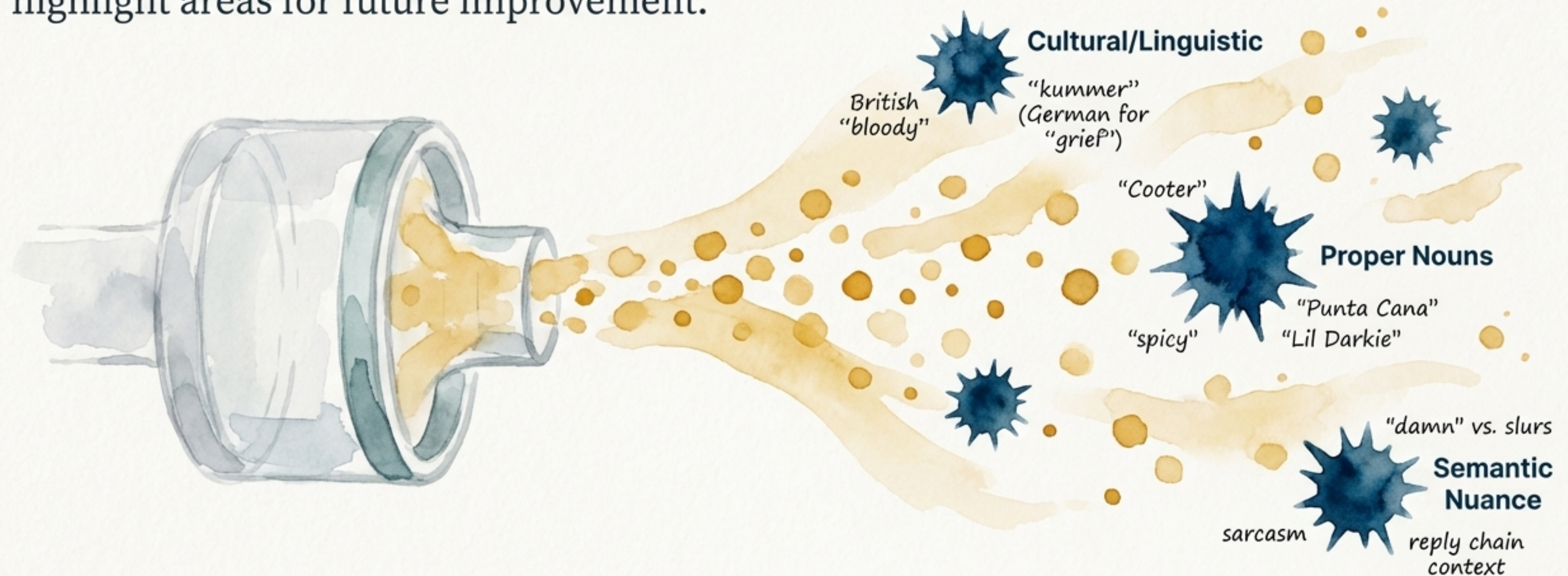
Precision (Accuracy of positive predictions):
High precision maintains user trust. The system acts as a reliable first-line filter.



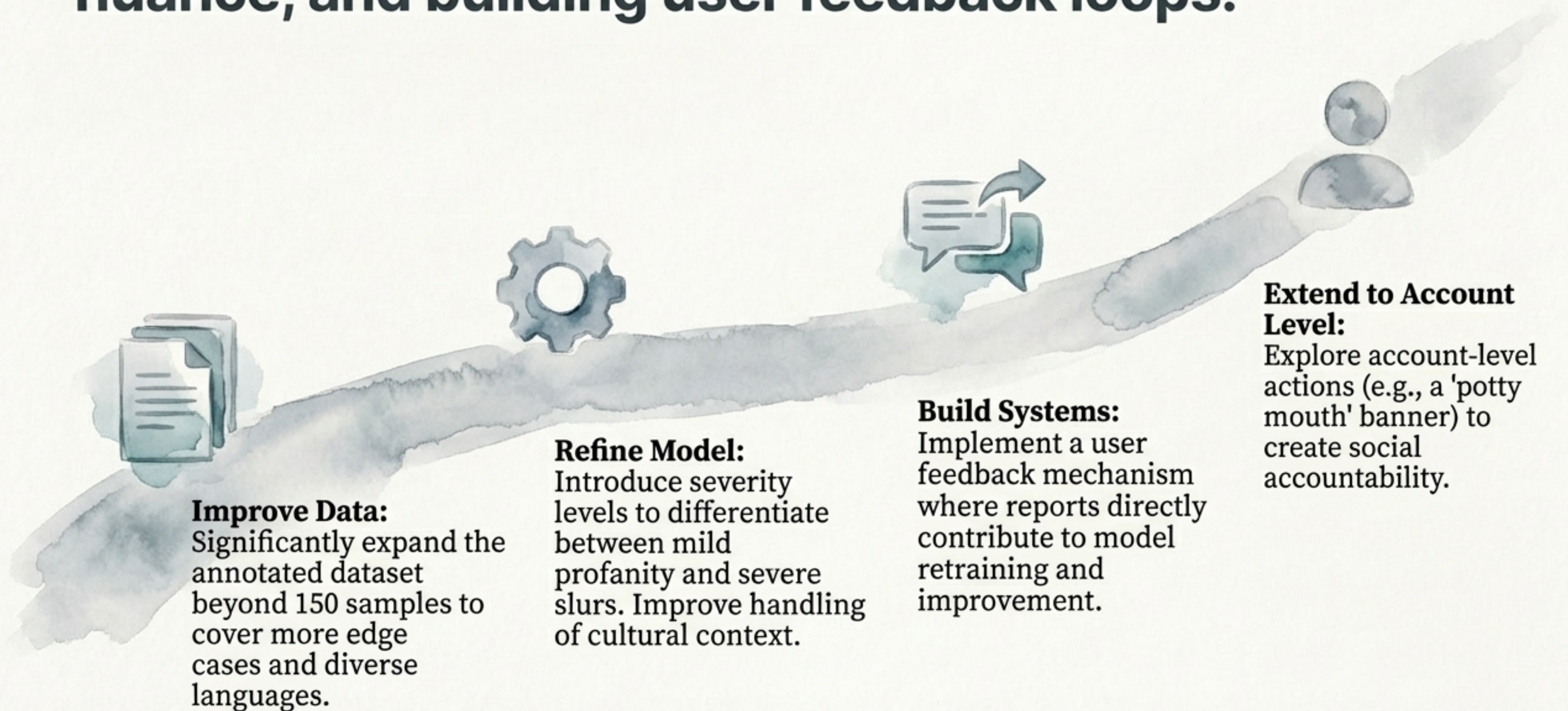
Recall (Ability to find all positives):
Slightly lower recall is an acceptable trade-off. Borderline cases can be reviewed by users and moderators.

The nuances of language and context reveal the system's current limitations.

We encountered numerous edge cases that highlight areas for future improvement.



Future work will focus on expanding data, refining nuance, and building user feedback loops.



The most ethical application is not as a gatekeeper, but as a tool for user empowerment.

Automated systems must be handled with care, acknowledging potential for misuse and the challenges of linguistic diversity. Our labeler is designed to give individual users control over their own feed, not to enforce a single, platform-wide standard. This approach respects both user safety and expressive freedom.

